A Hierarchical ReAct Multi-Agent Pipeline for Residential Appliance Load Scheduling Under Dynamic Electricity Pricing

Can language model agents give households reliable, cost-saving daily appliance recommendations?

Nguyen Hoan Le · Master's thesis, Information Processing Science · University of Oulu

Contents

01	**Introduction**	Motivation, previous research, research questions
02	**Artifact and method**	The two-agent pipeline, data and the pre-registered evaluation
03	**Demonstration**	The demo application in screenshots
04	**Results**	SQ1 value and cost · SQ2 conditions · SQ3 one agent or two
05	**Conclusion**	Significance, limitations and future work

Finnish prices change every 15 minutes

15 min

price interval for spot-price contracts since Oct 2025

3.1×

median ratio of the day's dearest to cheapest hour, 2023–2026

1 in 5

days with at least one negative-price hour

8 kW

power-charge threshold; a car and a sauna come close to it

The problem Fitting several appliance cycles to a new price curve every day, under a power limit, is hard by hand. Could an AI assistant do it for any household, in plain language?

LLM energy assistants exist, but are not yet reliable

What earlier LLM systems showed

- Plans for several appliances often failed
- Tested on very few days
- Rules sometimes broken
- Never compared with existing tools

What this thesis adds

- An optimiser tool and a verifying harness around the agents
- Evaluation over many price patterns and loads, on held-out days
- Comparison with tools households can already install

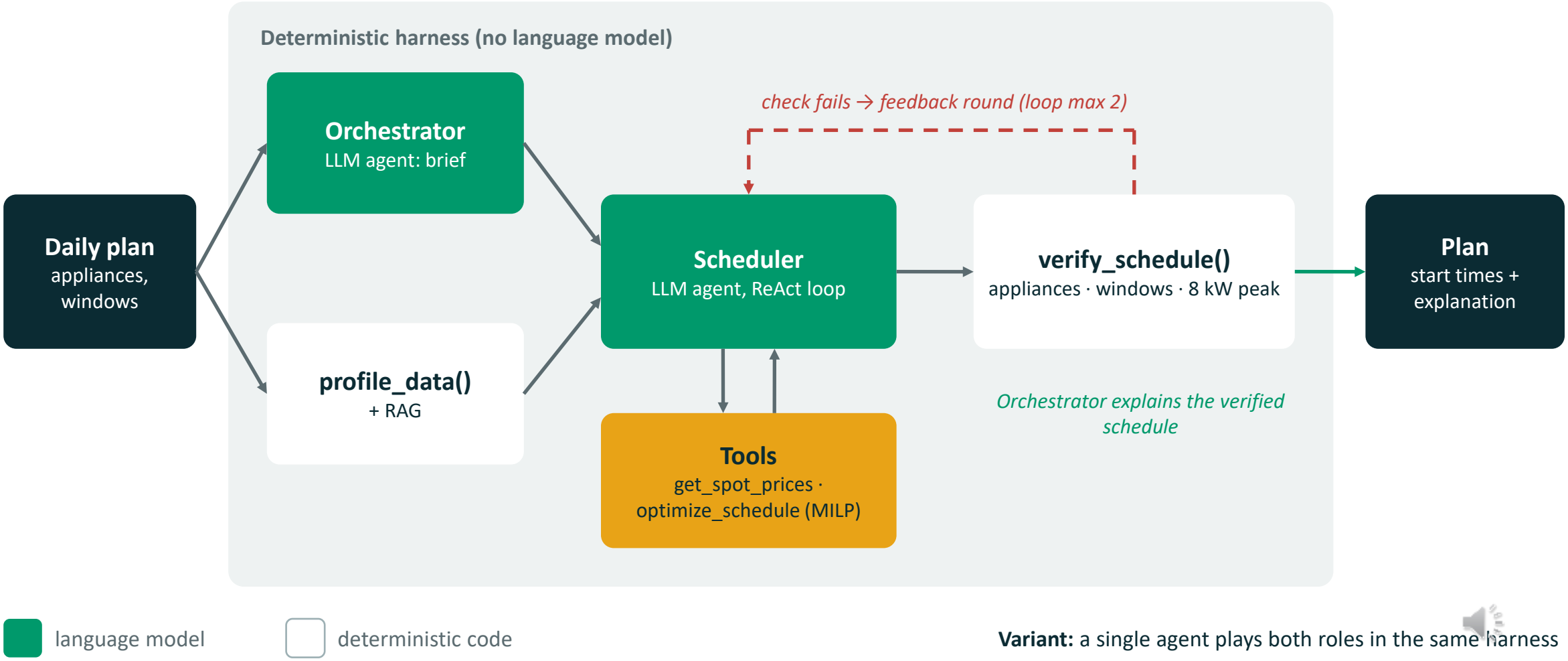
RQ How can a hierarchical ReAct multi-agent system (the Multi-Agent Pipeline) be designed to give households reliable, cost-saving daily appliance recommendations under dynamic electricity prices?

SQ1 What do the pipeline's recommendations save, how often are they valid, and what do they cost, compared with a base LLM without tools?

SQ2 Under which price patterns and household loads do they save at least as much as existing home energy management tools?

SQ3 Does dividing the work between two agents make them more reliable, cheaper or faster than a single agent?

The artifact: two agents inside a deterministic harness



Variant: a single agent plays both roles in the same harness

Evaluation on 72 held-out days, pre-registered

Data

- REFIT: real appliance cycles from 18 UK homes
- Finnish Nord Pool prices, every 15-minutes
- 4 price patterns × 3 loads
- Electrified: +3.7 kW car, +6.8 kW sauna

Design of the test

- 9 hypotheses registered before the run
- Held-out set Q3: 72 days, unseen in design
- GPT-6 Luna, 3 runs per day
- Wilcoxon, TOST and Mann-Whitney tests.

Compared with

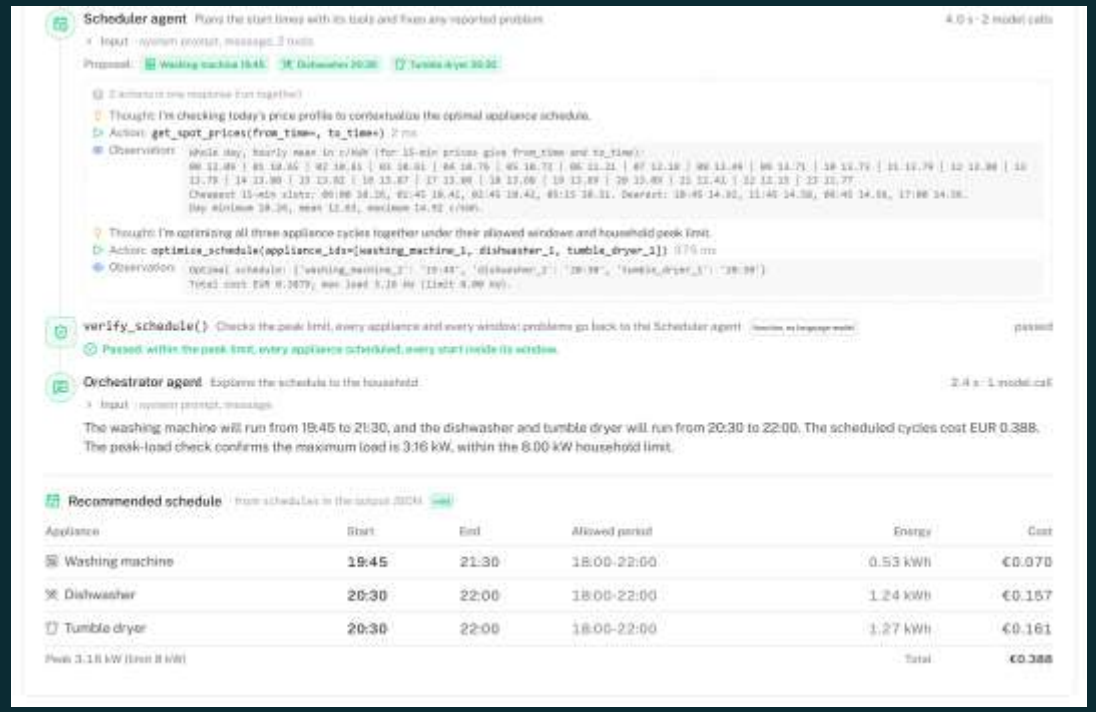
- Exact optimum (MILP), upper bound
- LLM without tools
- Cheapest window; HA cheapest hours
- EMHASS optimiser; smart timer

Primary outcome saving against starting every appliance when ready; a plan that breaks a window or the 8 kW limit is invalid

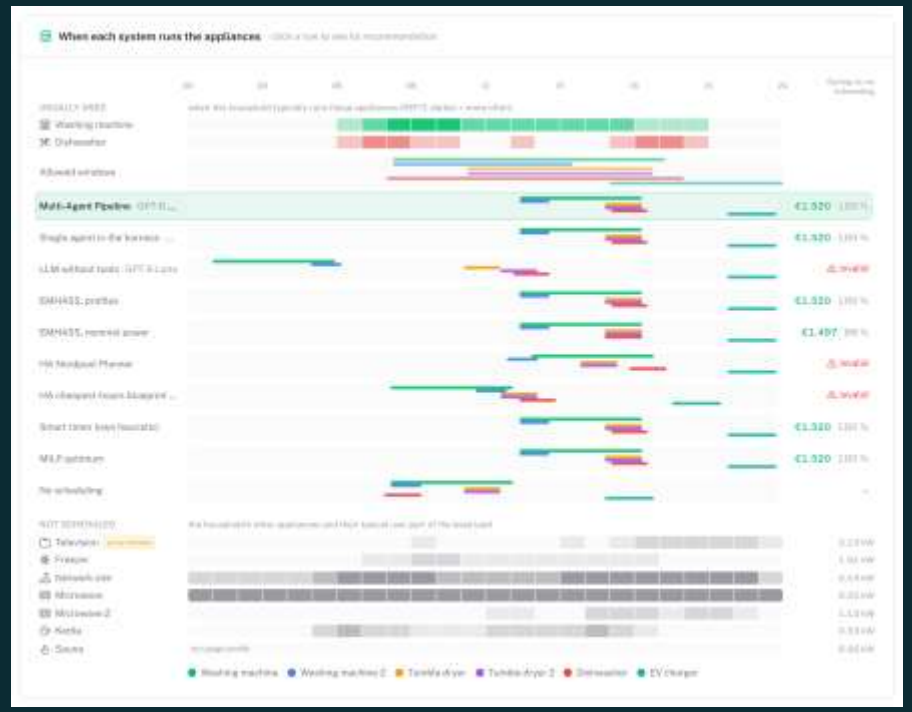
REFIT: Personalized Retrofit Decision Support Tools for UK Homes using Smart Home Technology
EMHASS: Energy Management for Home Assistant

The demo application

Try it: a live run, verified in under 10 s



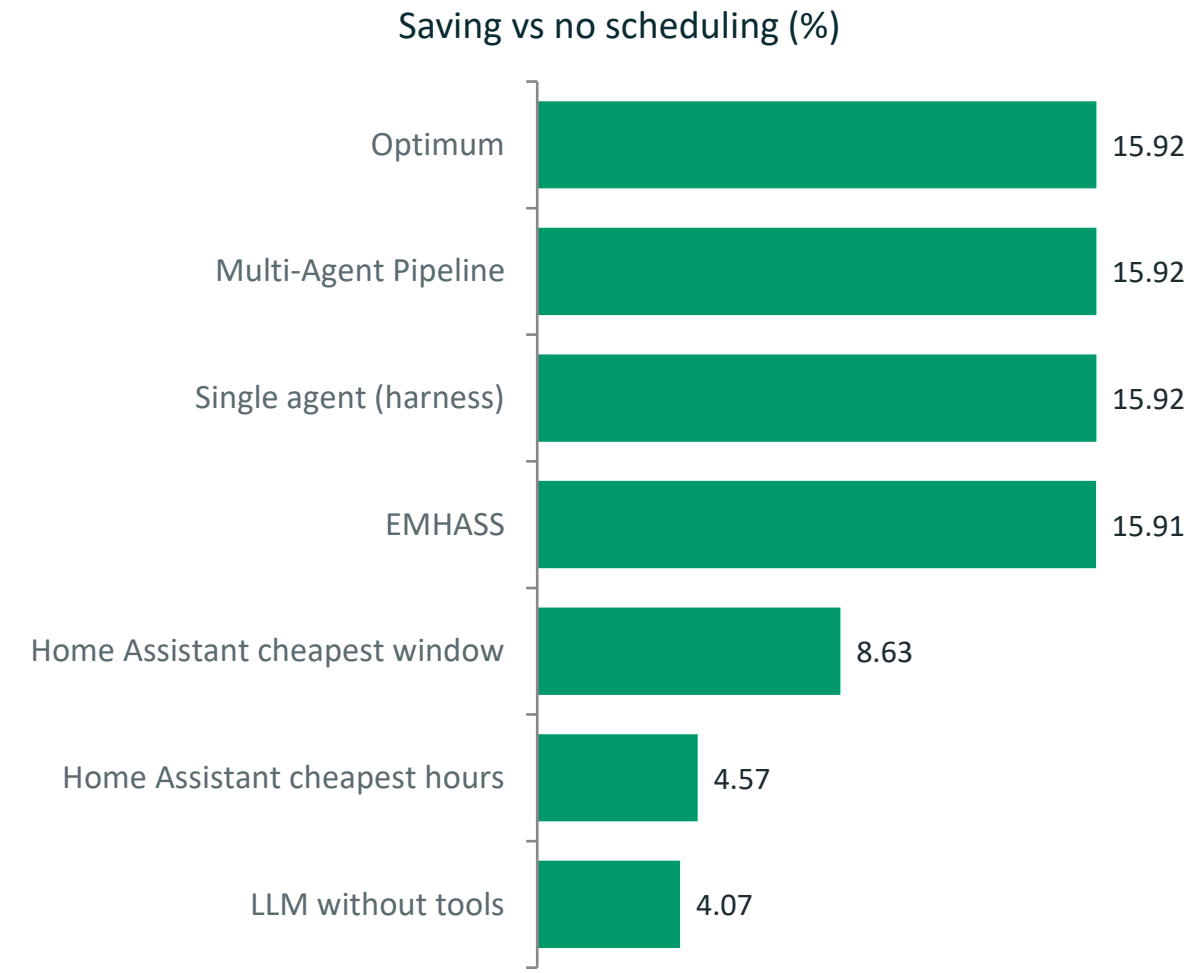
Replay: one held-out day, every system

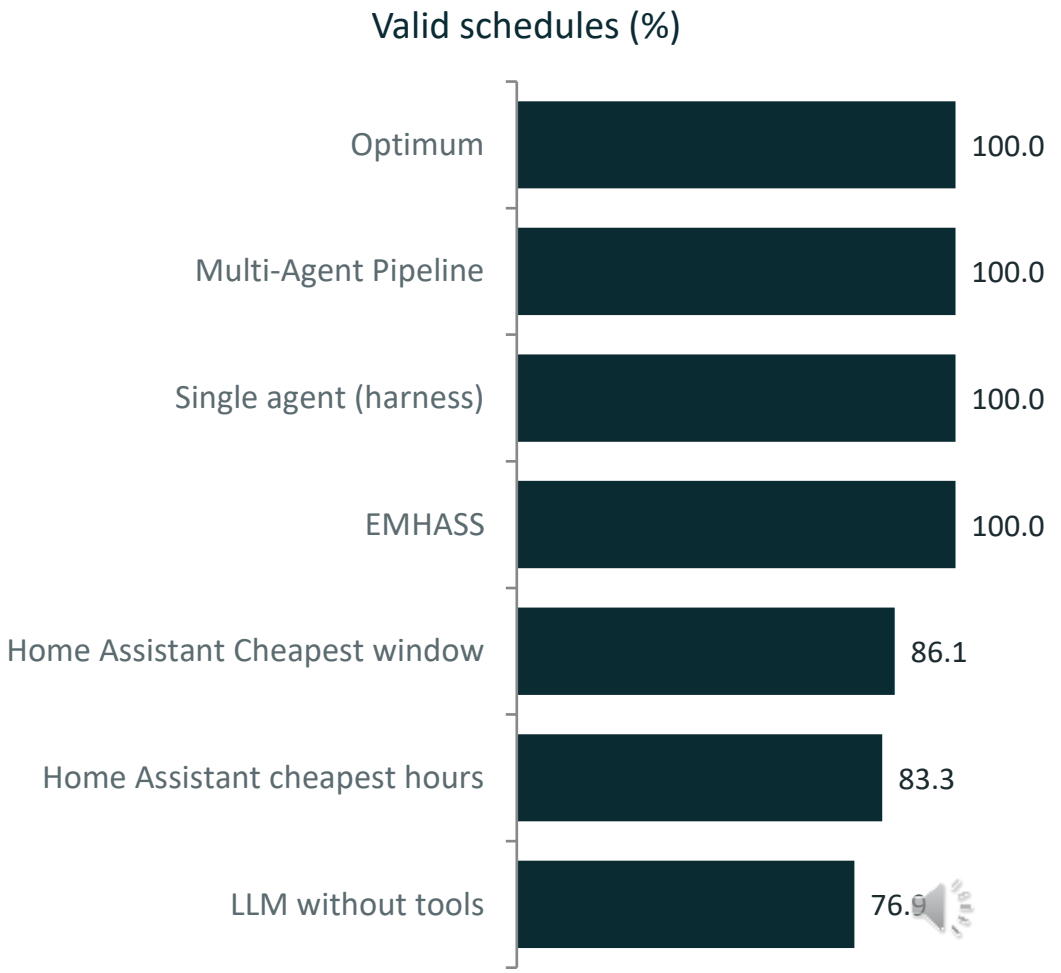


hems-demo.lenguyenhanhoan.workers.dev · GPT-6 Luna · today's Finnish prices from ENTSO-E

Results:
Same model, different design: 4 % → 16 % saving, all valid

Valid: every appliance runs inside its allowed time window and the household never goes over the 8 kW limit.





Held-out set Q3, 72 days; LLM systems: GPT-6 Luna, 3 runs per day (216 runs each)

Fast and cheap enough for a daily recommendation



7.4 s

Pipeline, mean response

objective: under 60 s



$0.0004

Pipeline, API cost per run

objective: under $0.10



5.0 s

Single agent: $0.00025

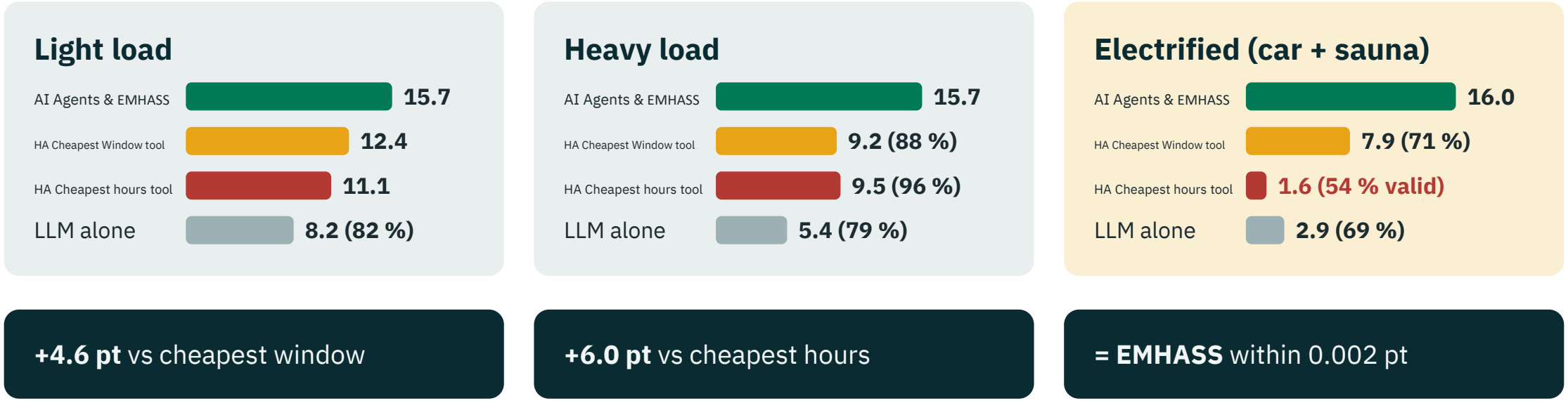
same schedules, faster and cheaper



Open model on a laptop gpt-oss-20b (local, no API) was valid and optimal in all 144 runs of each design on the exploratory benchmark; median 22–34 s.



Rules fall behind as the load grows; EMHASS keeps up



+4.6 pt vs cheapest window

+6.0 pt vs cheapest hours

= EMHASS within 0.002 pt

Saving vs no scheduling (%), pooled over Q3; in brackets the share of valid plans where below 100 %. Pipeline = single agent = EMHASS = optimum.

7 of 9 pre-registered hypotheses supported

	Hypothesis	Estimate	p (Holm)	
H1a	Pipeline saves more than the LLM without tools	+7.40 pt	< 0.001	**supported**
H1b	Pipeline is valid more often than the LLM without tools	+23.1 pt	< 0.001	**supported**
H1c	Mean response time below 60 s	7.4 s	< 0.001	**supported**
H1c	Mean API cost below $0.10	$0.0004	< 0.001	**supported**
H2a	Pipeline saves more than the cheapest window	+4.65 pt	< 0.001	**supported**
H2b	Pipeline saves more than HA cheapest hours	+5.99 pt	< 0.001	**supported**
H2c	Pipeline equals EMHASS within ±1 pt (TOST)	0.002 pt	< 0.001	**supported**
H2d	Advantage over cheapest hours larger on electrified days	+6.66 pt	0.30	**not supported**
H2e	Advantage over cheapest window larger on electrified days	+2.40 pt	0.51	**not supported**



SQ1: H1a–c, SQ2: H2a–e. N = 72 held-out days; Holm adjustment within each sub-question; analysed exactly as registered.

SQ3: reliability comes from the harness, not a second agent

One agent or two?

- Same schedules in every run
- Single agent faster: 5.0 s vs 7.4 s
- Cheaper: $0.00025 vs $0.00040
- Feedback round never needed on Q3

Differences significant (Wilcoxon, $p < 0.001$); not pre-registered, reported descriptively

Design principles (4 of 6)

Let a solver search

Numbers and the combinatorial search come from deterministic tools

Verify with code

Every schedule is checked by code and returned when it fails

Fixed work in the harness

Profiling and retrieval are steps, not agents

Add an agent only with decisions of its own

The second agent only restated what the harness provided

Conclusion: What it means, and what it does not show



Significance

- An LLM alone is not a reliable scheduler; the same model in the right design is valid, optimal and cheap
- The agent's value is access, not better optimisation: a plain description in, an explained and verified plan out
- A reusable pattern for agentic systems with hard constraints



Limitations

- UK appliance data, Finnish prices
- Simulated households; recommends only, no device control
- Structured requests, not free conversation
- 8 kW applied as a hard limit per 15-minute slot
- Modest savings in euros: at most about €35 a year per household

Takeaway the design, not the model, makes LLM energy advice reliable



One or More - When would a second agent pay off?

Practitioner criteria for several agents

Microsoft (2025)

Security or compliance boundary	no
Several teams or domains	no
More than 3 to 5 functions	no
A critic reviewing output	by code
Too many tools for one agent	no
Subtasks in parallel	by code
Different personas or guardrails	no

Next step: from advice to action



Let the agents decide and explain, a solver search, and code verify

RQ

Agents in a deterministic harness with an optimiser tool; six design principles

SQ1

Valid and optimal in every run; 7.4 s and $0.0004 per recommendation

SQ2

Beats the Home Assistant rules under every pattern and load; equals EMHASS

SQ3

A second agent adds time and cost, not reliability; one agent suffices

Future work field trial · a device control agent for smart devices · free-text requests · smaller local models



Thank you · Hoan Le · University of Oulu